



2019-A

1.  $f(z) = \frac{1}{2i} \left( \frac{z}{z} - \frac{\bar{z}}{z} \right)$  ( $z \neq 0$ ) 证当  $z \rightarrow 0$   $f(z)$  极限不存在

$$z = Re^{i\theta}$$

$$f(z) = \frac{1}{2i} (e^{i2\theta} - e^{-i2\theta}) = \frac{2i \sin 2\theta}{2i} = \sin 2\theta$$

当  $z \rightarrow 0$  即  $R \rightarrow 0$  时,  $\sin 2\theta$  极限不存在

$\therefore \lim_{z \rightarrow 0} f(z)$  极限不存在

2.  $w = \frac{1+i}{z}$   $x^2 + y^2 = 4$  映射为?

$$z = x + iy$$

$$w = u + iv = \frac{1+i}{x+iy} = \frac{(1+i)(x-iy)}{x^2+y^2} = \frac{1+i}{4} (x-iy)$$

$$= \frac{x+y}{4} + i \frac{x-y}{4}$$

$$u^2 + v^2 = \frac{(x+y)^2 + (x-y)^2}{16} = \frac{2x^2 + 2y^2}{16} = \frac{1}{2}$$

$$\text{即 } u^2 + v^2 = \frac{1}{2}$$

3.  $f(z) = x^3 - y^3 + i 2x^2y^2$

$$u = x^3 - y^3$$

$$v = 2x^2y^2$$

$$\frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial y} \therefore \text{不解析.}$$

$$\frac{\partial u}{\partial x} = 3x^2$$

$$\frac{\partial v}{\partial y} = 4x^2y$$

$$4. \oint_{|z|=1} \frac{|dz|}{|z-2|^2} \quad \tilde{z} z = e^{i\theta} = \cos\theta + i \sin\theta$$

$$|dz| = |d\theta|$$

$$|z-2|^2 = |\cos\theta - 2 + i \sin\theta|^2 = 5 - 4\cos\theta$$

$$I = \int_0^{2\pi} \frac{d\theta}{5 - 4\cos\theta}$$

$$= \oint_{|z|=1} \frac{1}{5 - 4 \cdot \frac{z+z^{-1}}{2}} \cdot \frac{1}{iz} dz$$

$$= \oint_{|z|=1} \frac{i}{2z^2 - 5z + 2} dz$$

$$= i \oint_{|z|=1} \frac{-\frac{3}{2}}{2z-1} + \frac{\frac{1}{2}}{z-2} dz$$

$$= \frac{2}{3}\pi$$

$$z = e^{i\theta}$$

$$dz = ie^{i\theta} d\theta = iz d\theta$$

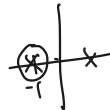
$$\frac{1}{iz} dz = d\theta$$

$$z = e^{i\theta} = \cos\theta + i\sin\theta$$

$$z^{-1} = e^{-i\theta} = \cos\theta - i\sin\theta$$

$$\cos\theta = \frac{z+z^{-1}}{2}$$

$$\sin\theta = \frac{z-z^{-1}}{2i}$$

$$5. I = \int_{C_j} \frac{z}{(z+1)(z-1)^2} dz \quad (j=1,2,3) \quad (z \in \mathbb{C})$$


1)  $C_1: |z+1| = \frac{1}{2}$     2)  $C_2: |z-1| = \frac{1}{2}$     3)  $|z|=2$

$$I_1 = 2\pi i \operatorname{Res} \left[ \frac{z}{(z+1)(z-1)^2}, -1 \right]$$

$$= 2\pi i \left. \frac{z}{(z-1)^2} \right|_{z=-1} = \frac{-\pi i}{2}$$

$$I_2 = 2\pi i \operatorname{Res} \left[ \frac{z}{(z+1)(z-1)^2}, 1 \right]$$

$$= 2\pi i \lim_{z \rightarrow 1} \frac{d\left(\frac{z}{z+1}\right)}{dz} = 2\pi i \left. \frac{z+1-z}{(z+1)^2} \right|_{z=1} = \frac{\pi i}{2}$$

$$I_3 = 2\pi i \operatorname{Res} \left[ \frac{z}{(z+1)(z-1)^2}, -1 \right] + 2\pi i \operatorname{Res} \left[ \frac{z}{(z+1)(z-1)^2}, 1 \right] = 0$$

6.  $f(z)$  在  $|z| < R$  上解析

$$\frac{R^2 - |z|^2}{2\pi i} \int_{|\eta|=R} \frac{f(\eta)}{(\eta - z)(R^2 - \bar{z}\eta)} d\eta = f(z)$$

$$|z| < R$$

$$|\eta| = R \text{ 逆时针}$$

$\therefore$  在  $|\eta| = R$  上积分  $\eta = z$  为奇点

$$|\bar{z} \cdot \eta| = |\bar{z}| \cdot |\eta| < R \cdot R = R^2$$

$\therefore$  仅有  $\eta = z$  奇点

$$\text{左式} = \frac{R^2 - |z|^2}{2\pi i} \cdot 2\pi i \cdot \text{Res} \left[ \frac{f(\eta)}{(\eta - z)(R^2 - \bar{z}\eta)}, z \right]$$

$$= \frac{R^2 - |z|^2}{2\pi i} \cdot 2\pi i \cdot \frac{f(z)}{R^2 - \bar{z} \cdot z} = f(z) = \text{右式}$$

7.  $u = 2(x-1)y$   $f(z) = -i$  求  $f(z) = u + iv$

$$a = u'_x = 2y \quad 2xy - 2y$$

$$b = -u'_y = 2(1-x)$$

$$f'(z) = a + ib = 2y + i 2(1-x)$$

$$= 2i - 2i z = -2i(z-1)$$

$$f(z) = -i(z-1)^2 + C$$

$$f(z) = -i \quad C = 0$$

$$f(z) = -i(z-1)^2$$

8.  $f(z) = \frac{z^5}{z^2 - 5z + 6}$   $z=0$  泰勒展开及收敛半径

$$f(z) = z^5 \left( \frac{1}{z-3} - \frac{1}{z-2} \right) \quad \text{收敛半径为2.}$$

$$= z^5 \left( \frac{-\frac{1}{3}}{1-\frac{z}{3}} + \frac{\frac{1}{2}}{1-\frac{z}{2}} \right)$$

$$= -\frac{1}{3} z^5 \sum_{n=0}^{\infty} \left(\frac{z}{3}\right)^n + \frac{1}{2} z^5 \sum_{n=0}^{\infty} \left(\frac{z}{2}\right)^n$$

$$= -\sum_{n=0}^{\infty} \frac{z^{n+5}}{3^{n+1}} + \sum_{n=0}^{\infty} \frac{z^{n+5}}{2^{n+1}}$$

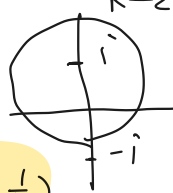
$$(2i)^{-2} = \frac{-1}{4}$$

$$R=2$$

9.  $f(z) = \frac{1}{(z^2+1)^2}$   $z=i$  的去心邻域展开洛朗  $\int_{|z|=1} f(z) dz$

$$R=2$$

$$f(z) = \frac{1}{[(z-i)(z+i)]^2} = (z-i)^{-2} (z+i)^{-2}$$



$$= (z-i)^{-2} (z-i + 2i)^{-2} = (z-i)^{-2} \left( \frac{z-i}{2i} + 1 \right)^2 \left( -\frac{1}{4} \right)$$

$$= -\frac{1}{4} (z-i)^{-2} \left[ 1 + \sum_{n=1}^{\infty} \frac{(-2)(-3)\cdots(-1-n)}{n!} \left( \frac{z-i}{2i} \right)^n \right]$$

$$= -\frac{1}{4} (z-i)^{-2} \left[ 1 + \sum_{n=1}^{\infty} \frac{(n+1)!}{n!} (-1)^n \frac{(z-i)^n}{2^n (i)^n} \right]$$

$$= -\frac{1}{4} (z-i)^{-2} - \sum_{n=1}^{\infty} i^n (n+1) \frac{(z-i)^{n-2}}{2^{n+2}} = -\sum_{n=0}^{\infty} i^n (n+1) \frac{(z-i)^{n-2}}{2^{n+2}}$$

$$C_{-1} = -\frac{i^n (n+1)}{2^{n+2}} \Big|_{n=1} = -\frac{2i}{8} = -\frac{i}{4}$$

$$\int_{|z|=1} f(z) dz = 2\pi i C_{-1} = \frac{\pi}{2}$$

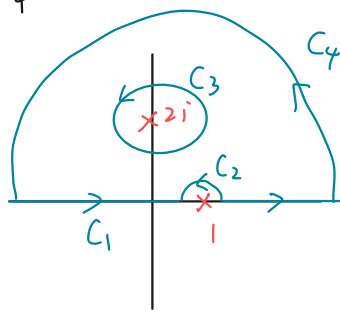
$$10. \quad I = \int_{|z|=3} \frac{(z^2-2)(z^3-1) \cos \left[ \frac{1}{(z-1)^2} \right]}{(2-z^5)(z-4)} dz$$

$$= \int_{|z|=\frac{1}{3}} \frac{(z^{-2}-2)(z^{-3}-1) \cos \left[ \frac{1}{(z^{-1}-1)^2} \right]}{(2-z^{-5})(z^{-1}-4)} z^{-2} dz$$

$$= \int_{|z|=\frac{1}{3}} \frac{(1-2z^2)(1-z^3) \cos \frac{z^2}{(1-z)^2}}{(2z^5-1)(1-4z)z} dz$$

$$= 2\pi i \frac{(1-2z^2)(1-z^3) \cos \frac{z^2}{(1-z)^2}}{(2z^5-1)z} \bigg|_{z=\frac{1}{4}} + 2\pi i \frac{(1-2z^2)(1-z^3) \cos \frac{z^2}{(1-z)^2}}{(2z^5-1)(1-4z)} \bigg|_{z=0}$$

$$11. \quad \int_{-\infty}^{+\infty} \frac{\sin x}{(x^2+4)(x-1)} dx$$



$$I_1 - I_2 + I_4 - I_3 = 0$$

$$I_1 = I_2 + I_3$$

构造  $I_1 = \int_{-\infty}^{\infty} \frac{e^{iz}}{(z^2+4)(z-1)} dz$

$$I_3 = 2\pi i \frac{e^{iz}}{(z+2i)(z-1)} \bigg|_{z=2i} = 2\pi i \frac{e^{-2}}{4i(2i-1)} = \frac{\pi e^{-2}}{2(2i-1)} = \frac{\pi e^{-2}(2i+1)}{-10}$$

//  $C_3$  积分用留数定理.

先观察  $\lim_{r \rightarrow 0} \int_{C_2} \frac{e^{iz}}{(z^2+4)(z-1)} dz$

step 1,  $\varphi(z) = \frac{e^{iz}}{z^2+4}$ ,  $\varphi(z)$  在  $z=1$  邻域内解析.

将  $\varphi(z)$  展开

$$\varphi(z) = \varphi(1) + \sum_{n=1}^{\infty} \frac{\varphi^{(n)}(1)}{n!} (z-1)^n$$

$$= \frac{e^i}{5} + \boxed{\varphi_1(z)}$$

// 后续证明这一块积分为 0

step 2.  $I_2 = \lim_{r \rightarrow 0} \int_{C_2} \frac{e^{iz}}{(z^2+4)(z-1)} dz$

$$= \lim_{r \rightarrow 0} \int_{C_2} \frac{e^i}{5(z-1)} dz + \boxed{\lim_{r \rightarrow 0} \int_{C_2} \varphi_1(z) dz} = 0$$

$$0 \leq \left| \lim_{r \rightarrow 0} \int_{C_2} \varphi_1(z) dz \right| \leq \lim_{r \rightarrow 0} \boxed{M} \pi r = 0$$

$\downarrow |\varphi_1(z)|_{\max}$   
 $\downarrow$   
 半个圆周

step 3. 变量代换

$$z = 1 + re^{i\theta}$$

$$dz = ire^{i\theta} d\theta$$

$$= \lim_{r \rightarrow 0} \int_0^\pi \frac{e^i}{5re^{i\theta}} ire^{i\theta} d\theta$$

$$= \int_0^\pi \frac{ie^i}{5} d\theta = \frac{i\pi}{5} (\cos(1) + i\sin(1)) = \frac{\pi}{5} (-\sin(1) + i\cos(1))$$

$$I = \operatorname{Im} [I_2 + I_3] = \operatorname{Im} \left[ \frac{\pi e^{-2}(2i+1)}{-10} - \frac{\pi}{5} \sin(1) + i \frac{\pi}{5} \cos(1) \right] = -\frac{\pi}{5} e^{-2} + \frac{\pi}{5} \cos(1)$$